

9.6.2 User Management

A tabular view of all registered users with columns for name, email, role, registration date, and verification status. Admin actions include:

- Edit user role (change between farmer, agent, admin)
- View user activity history
- Filter users by role using tab-based navigation

9.6.3 Listing Oversight

A complete view of all platform listings across all users. Admin-specific actions include:

- Mark DELIVERED listings as “Recycled” to complete the waste lifecycle
- View listing details including photos, location, and assignment history
- Filter listings by status using dropdown selectors

9.6.4 Analytics Dashboard

Interactive charts built with the Recharts library provide visual insights:

- **Waste by Type:** Pie chart showing distribution of waste categories across all listings
- **Collection Over Time:** Line/bar chart showing listing creation and completion trends over time
- **Status Distribution:** Bar chart comparing listings across all status categories
- Charts are responsive and include tooltips for detailed data inspection

9.7 Payment Processing Flow

The Razorpay integration provides a complete payment workflow:

1. The `PaymentButton` component appears on DELIVERED listings
2. Clicking the button triggers a server-side API call to create a Razorpay order
3. The Razorpay checkout modal opens with pre-filled amount and merchant details
4. The modal supports multiple payment methods: UPI (scan QR or enter UPI ID), debit/-credit cards, net banking, and wallets
5. Upon successful payment, a success toast is displayed and the listing’s payment status is updated
6. Payment confirmations are stored in the orders collection for audit purposes